Լաբորատոր աշխատանք 9

Աշխատանքի նպատակը.

Խնդրի պահանջը. *y' = ln(x + 1) + y / (x + 1)* Հաշվել լուծման 21 արժեք և գտնել վերջնական ու սկզբնական արժեքների տարբերությունը:

Տեսական մաս.

```
#include <iostream>
#include <cmath>
using namespace std;
class Kosh {
private:
    int size;
    double* x;
    double* y;
public:
    Kosh(double x0, double y0) {
        size = 21;
        double h = 0.05;
        x = new double[size];
        y = new double[size];
        double ln_x0 = log(x0 + 1);
        double c = (y0 / (x0 + 1)) - (ln_x0 * ln_x0) / 2.0;
        for (int i = 0; i < size; ++i) {
            x[i] = x0 + i * h;
            double ln_xi = log(x[i] + 1);
            y[i] = (x[i] + 1) * ((ln_xi * ln_xi) / 2.0 + c);
        }
    }
    Kosh(const Kosh& other) {
        size = other.size;
        x = new double[size];
        y = new double[size];
        for (int i = 0; i < size; ++i) {
            x[i] = other.x[i];
            y[i] = other.y[i];
        }
    }
    Kosh& operator=(const Kosh& other) {
        if (this == &other) {
            return *this;
        }
        delete[] x;
        delete[] y;
        size = other.size;
        x = new double[size];
        y = new double[size];
        for (int i = 0; i < size; ++i) {
            x[i] = other.x[i];
            y[i] = other.y[i];
        }
        return *this;
    }
    ~Kosh() {
        delete[] x;
```

```
        delete[] y;
    }
    void show() const {
        cout << "   x[i]          y[i]" << endl;
        cout << "-------------------------" << endl;
        for (int i = 0; i < size; ++i) {
            cout << x[i] << "      " << y[i] << endl;
        }
        double diff = y[size - 1] - y[0];
        cout << "\nInitial value (y[0]): " << y[0] << endl;
        cout << "Final value (y[" << size - 1 << "]): " << y[size - 1] << endl;
        cout << "Difference (final - initial): " << diff << endl;
    }
};
int main() {
    double x0, y0;
    cout << "Enter x0 and y0 values (e.g., 0  1): ";
    cin >> x0 >> y0;
    cout << "\n--- First object (k1) ---" << endl;
    Kosh k1(x0, y0);
    k1.show();
    cout << "\n--- Second object (k2 copied from k1) ---" << endl;
    Kosh k2 = k1;
    k2.show();
    return 0;
}
```